

BLOCK 2

MODERN THEORIES OF INTERNATIONAL TRADE

Unit 3	Intra-Industry Trade
Unit 4	Alternative Explanations of Trade

BLOCK INTRODUCTION

This block presents the modern theories of International Trade. It discusses the need, rational and measurement of Intra-Industry Trade (IIT). This block also elucidates the extensions of classical trade theories with more realistic assumptions which explain the current patterns of international trade in the world market. This block comprises of two units.

Unit 3 Intra-Industry Trade (IIT) deliberates on the phenomena of intra-industry trade and identifies the need for a different theory for intra-industry trade. The two types of IIT – horizontal and vertical IIT are also elaborated in this Unit. The unit also explained ‘love of variety’ and ‘ideal variety’ approach in the Intra-Industry trade. Possible reasons behind IIT in identical products are also discussed in this unit. The unit also presents the Grubel Lloyd (GL) index to measure the intensity of IIT.

Unit 4 Alternative Explanations of International Trade Theories sheds light on new, complementary trade theories based on real world situations using the concepts of Economies of Scale, Imperfect Competition, Product differentiation and Technological Gap. This unit probe the effect of relaxing each of the classical and Heckscher-Ohlin trade theory assumptions on International trade. It critically examines the impact of Logistics cost and environmental standards on the world trade.

UNIT 3 INTRA-INDUSTRY TRADE *

Structure

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3.0 OBJECTIVES

After studying this Unit, you should be able to:

- describe the meaning of Intra-Industry trade;
- distinguish between horizontal and vertical Intra-Industry trade;
- identify the need for a separate theory for Intra-Industry trade;
- explain the 'love of variety' and 'ideal variety' approach in Intra-Industry trade;
- elucidate Intra-Industry trade in Horizontally and vertically differentiated Commodities;
- list the major problems in measurement of intensity of Intra-Industry trade; and
- measure intensity of Intra-Industry trade using the Grubel Lloyd (GL) index.

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3.1 INTRODUCTION

We have already discussed in the previous unit about the gains from international trade i.e., the various benefits a country derives from trading goods and services with other countries. Each trading country gains when the total output increases due to the division of labour and specialisation. This unit will discuss the phenomena of intra-industry trade and why there is a need for a different theory for intra-industry trade. Greater specialisation in the manufacturing of unique varieties or product lines by individual plants in different countries has led to the tremendous growth of IIT. In order to exploit economies of scale in production, individual plants specialize in unique variety and thus leading to growth of IIT.

Further, greater specialization at the level of distinct components used in an industry has promoted IIT in intermediate goods. In this unit, we will also discuss about the two types of IIT – horizontal and vertical IIT. Horizontal IIT is the simultaneous export and import of differentiated products within an industry of a similar quality. Vertical IIT, on the other hand, refers to the exchange of differentiated products of different qualities.

We will also discuss ‘love of variety’ and ‘ideal variety’ approach in the Intra-Industry trade. We will further discuss the possible reasons behind IIT in identical products.

Two major issues in the empirical analysis of IIT are concerned with: i) the definition of ‘industry’ and ii) measurement of IIT. We will discuss about these issues along with the Grubel Lloyd (GL) index to measure the intensity of IIT.

3.2 TRADE LIBERALIZATION AND THE PHENOMENON OF INTRA-INDUSTRY TRADE

Mercantilism Trade liberalisation is expected to rationalise the choice of product lines by individual plants in domestic industries. Production and export of all varieties in a given industry become impossible if economies of scale in production are to be reaped. Suppose the products in an industry are differentiated and each is manufactured with increasing returns to scale. In that case, a country may specialise in producing a subset of varieties for home consumption and export, while importing those varieties that are not supplied domestically. In this context, it becomes important to understand the structure of exports in its relation to imports within the industry. For understanding this dimension of trade pattern, trade economists have evolved the concept of intra-industry trade (henceforth IIT), which means the **simultaneous occurrence of exports and imports within the same industry**. To be more precise, IIT means the simultaneous export and import of products, which are very close substitutes for each other in terms of factor inputs and consumption.

Growth of IIT is an outcome of greater specialisation in the manufacturing of unique varieties or product lines by the individual plants in different countries.

Because there are fixed costs associated with the production of each variety, it is imperative that each plant specializes in a unique variety if economies of scale in production are to be exploited. Therefore, the firms in each country would manufacture a subset of the varieties within an industry for meeting home demand and export. At the same time, the varieties that are not produced within the country will be imported to meet the domestic demand. This leads to the occurrence of IIT in final consumer goods. Further, greater specialization at the level of distinct components used in an industry will promote IIT in intermediate goods.

The phenomenon of IIT began to gain sustained interest since the early 1960s, after the formation of the European Economic Community (EEC). The conventional wisdom is that competition would induce a process of resource reallocation from the import competing industries to those industries where the country has comparative advantages. It follows that a natural consequence of trade liberalization is the expansion of inter industry trade – that is, export increase from one set of industries and import increase in another. Further, it is held that trade liberalization invariably involves adjustment costs as some domestic industries may go out of business.

However, a number of studies in the context of different countries suggest that trade liberalization biases the country's trade expansion towards IIT rather than inter industry trade. In other words, the pattern of specialisation was not the one whereby different countries specialised in different products. Instead, different countries tended to specialise in different types of a given product and there was an expansion of exports from practically every industry rather than the demise of inefficient manufacturing industries.

In a liberalized environment, different countries would specialise in different types of a given product and there could be expansion of exports from practically every industry. The apprehension that import liberalisation might lead to the demise of domestic industries in the developing countries (the fear of de-industrialisation) becomes untenable if we indeed find that the intensity of IIT increases with the reduction of trade barriers. The widely held 'smooth adjustment hypothesis' states that a high share IIT will be associated with relatively low adjustment costs, since, with intra-industry adjustment, workers move within industries rather than between them.

There is the distinction between the two types of IIT – horizontal and vertical IIT. Horizontal IIT refers to the simultaneous export and import of differentiated products within an industry of a similar quality. Vertical IIT, on the other hand, refers to the exchange of differentiated products that are of different qualities. Horizontal IIT arises when countries with similar factor endowment ratios specialize in unique varieties to exploit internal scale economies. Vertical IIT, on the other hand, arises when countries with different factor endowment ratios specialize in varieties that are differentiated by quality: capital-abundant

countries would specialize in the production of higher quality varieties while labor-abundant countries would specialize in low quality varieties. Vertical IIT between two countries will arise given an overlap in the demand for different qualities. This possibility is enhanced by the greater factor endowment difference between the countries. In practice, the trade that emerges from specialization at different stages of producing a particular good may also be treated as vertical IIT (for example, China imports parts of a product and exports the finished product). The extent of adjustment costs associated with trade liberalization can differ depending on whether IIT is horizontal or vertical. It is generally held that while any type of IIT entails relatively lower adjustment costs than inter-industry trade, the adjustment cost will be the least when IIT is horizontal in nature.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Describe Intra-Industry trade.

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2) Differentiate between horizontal and vertical IIT?

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3) Discuss the impact of trade liberalization on growth of IIT in a country.

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3.3 THEORY OF INTRA-INDUSTRY TRADE

3.3.1 Need for a Separate Theory for Intra-Industry trade

The early empirical studies on IIT stimulated analyses that were directed at establishing the extent of the phenomenon in different countries at different points in time. Grubel and Lloyd (1975) are noteworthy among these works, which documents detailed evidence of IIT for all major industrialised countries. While Grubel and Lloyd (1975) suggested a number of plausible explanations for IIT, it remained 'a phenomenon in search of a theory' till the late 1970s.

‘What determines the pattern of international trade?’ This question has been a subject of considerable interest to economists ever since the formulation of comparative advantage theory by David Ricardo. The well-known Heckscher-Ohlin-Samuelson (H-O-S) model, the synthesis of the initial attempts, remained the main theory for quite a long time. This model provides explanation for the emergence of inter-industry trade based on factor endowment difference of countries. This theory's core is the proposition that trade flows between countries are in ‘complementary’ goods with differing factor intensities. Accordingly, the bulk of empirical testing since then has been on exploring such complementarity.

Ever since Leontief’s results cast doubt upon the H-O-S explanations (the well-known Leontief paradox), there has been considerable work on refining the tests and restating the models to make them more suitable for testing. The scepticism on the explanatory power of the H-O-S model originated from the Leontief paradox apart, the new attempts have been stimulated by yet another observation. That is, a predominant amount of international trade takes place among industrial nations, with relatively similar resource endowments, which is difficult to reconcile with the prediction of the H-O-S model. It was generally held that the traditional trade models, based on the assumption of perfect competition cannot provide a proper understanding of IIT, particularly among the industrial countries.

Clearly, what appeared to be imperative was a general equilibrium theory of international trade, which incorporates the elements of economies of scale, product differentiation and imperfect competition. The pursuit of the task, however, was less easy than the stating of it. A major difficulty was that scale economies internal to firms are incompatible with perfect competition that was assumed in all traditional models. In short, theorists in the field lacked analytic tools necessary to formalise and refine the ideas in tractable mathematical models.

However, some major developments in economic theory in the 1970s completely changed the scenario. First, there was substantial progress in the theoretical modelling of imperfect competition, which led to the revival of Chamberlin's "large group" analysis of competition between similar firms producing differentiated products. Secondly, the later half of 1970s witnessed rigorous theoretical analysis of the process of product differentiation, by Spence (1976) and Dixit and Stiglitz (1977). This approach, called the "love-of variety approach" considers that each consumer has a taste for many different varieties. Later, Lancaster (1979) advanced an alternative approach, known as the "ideal variety approach", which considers that each consumer has a desire for the most preferred variety rather than many varieties. Though there is a significant difference in the two approaches, both represent cases of what is called "horizontal differentiation".

These developments provided economists with appropriate toolkit to facilitate the incorporation of product differentiation, economies of scale and imperfect competition into the formal trade theory and proliferation of theoretical research on IIT. The new approach proved useful for providing theoretically convincing explanations for IIT and integrating them with the conventional explanations for inter-industry trade based on factor proportions. It may, however, be noted that as long as a general theory of imperfect competition does not exist, it is not possible to arrive at a general theory of trade with imperfect competition. There are too many ways in which imperfect market structure can be modelled and in principle there can be as many models of specialisation and trade.

The theoretical literature may be grouped as belonging to four different strands. The earlier attempts represent models of horizontal IIT – that is, the exchange of commodities differentiated by those attributes excluding quality. Horizontal IIT is explained by economies of scale in the presence of product differentiation and imperfect competition. Such models have greater relevance for understanding the occurrence of IIT among developed countries. Significant attempts were made in the later years to offer a different strand of analysis, which represent cases of vertical IIT – that is, the exchange of commodities differentiated by quality. In general, these models explain the pattern of IIT along the lines similar to the pattern of inter-industry trade predicted in the conventional trade model, according the central role to factor endowment differences. Such models are considered to be particularly relevant to explain the presence of IIT between unequal trade partners. A third strand of analysis attempted to explain IIT in differentiated intermediate products in the presence of international returns to scale. Finally, some theoretical attempts have aimed at providing explanations for IIT in identical commodities on the basis of strategic interactions of internationally oligopolistic firms. In what follows, we shall sum up the essential elements of the major strands of theories.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What led to need for a Separate Theory for Intra-Industry trade?

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2) Describe ‘love of variety approach’ in intra industry trade.

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3) Describe 'ideal variety approach' in intra industry trade.

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3.3.2. IIT in Horizontally Differentiated Commodities.

Theoretical models of IIT in horizontally differentiated commodities are generally based on the Chamberlinian notion of monopolistic competition. As regards the treatment of product differentiation, the love-of-variety approach and the ideal variety approach are used by various economists. Notwithstanding differences in the approach followed to specify the demand side, all these models yielded broadly similar results. The essential arguments of these models may be summed up as follows.

In Krugman's model, the love of variety approach is assumed on the demand side. Each consumer has the same utility function, wherein all varieties enter symmetrically. The utility function for each of the (identical) consumer is:

$$U = \sum_{i=1}^N v(c_i), \quad v' > 0, \quad v'' < 0 \quad (1)$$

Where c_i is the consumption of each variety $i = 1, \dots, N$. Given the assumption of diminishing marginal utility from each variety ($v'' < 0$), aggregate utility (U) can be maximised by increasing the number of varieties (N) consumed rather than by spending the income on few varieties.

The economy under consideration is assumed to have only one factor of production - that is, labour. The underlying market structure is assumed to be the Chamberlinian monopolistic competition. A large number of firms exist, each of which produces a different variety (y_i) of the same good Y , but with the same cost function:

$$L_i = \alpha + \beta y_i \quad (2)$$

α stands for fixed labor input while β is marginal labor input. The term α incorporates economies of scale into the model. In this model, each firm produces a unique variety as no two firms have incentive to produce the same variety. This is due to the fact that economies of scale operate at the level of varieties and, at the same time, all available varieties enter the consumer's utility function.

The marginal cost, MC_i for the firm is $w\beta$ while the average cost is given as:

$$AC_i = \frac{wL_i}{y_i} = \frac{w(\alpha + \beta y_i)}{y_i} = \frac{w\alpha}{y_i} + w\beta \quad (3)$$

This means that AC falls with the increase in output (y_i) while the MC is constant (see Figure 3.1).

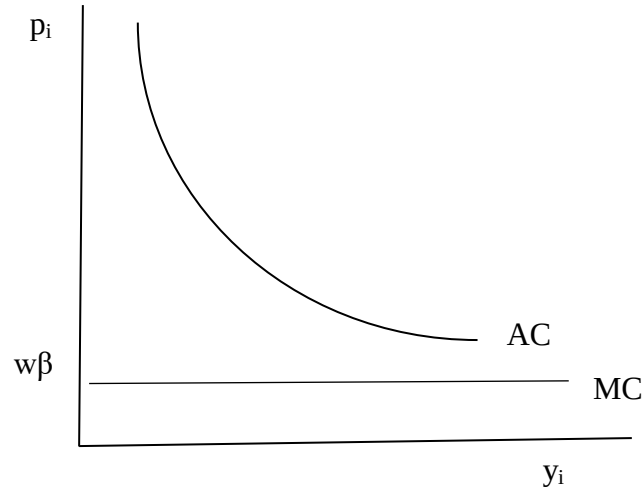


Figure 3.1: Average Cost Curve and Marginal Cost curve

The symmetry assumption in regard to both cost and demand functions will ensure that all varieties actually produced will be produced in the same quantity and at the same price. Thus, to arrive at equilibrium, one needs to determine only the price and output of a single representative variety (therefore, we can ignore the subscript i). The two equilibrium conditions for firms under monopolistic competition are: (i) $MR=MC$ and (ii) $p = AC$ (zero profit in the long run), where MR is marginal revenue and p is price. When η is the elasticity of demand for variety i , MR equals $\frac{p}{1} \left(1 - \frac{1}{\eta}\right)$

Using $MR=MC$ condition, we can show that:

$$\frac{p}{w} = \beta \left(\frac{\eta}{\eta - 1} \right) \quad (4)$$

Similarly, using the $p = AC$ condition, it can be shown that

$$\frac{p}{w} = \left(\frac{\alpha}{Lc} \right) + \beta \quad (5)$$

Where η is the elasticity of demand for variety i . We assume that $d\eta_i/dc_i < 0$ - that is, elasticity of demand decreases with increase in consumption of a variety (note that this assumption always hold for any linear demand curve, $q = a - bp$). With this assumption, it is clear that the graph of equation 4 (AB curve) slopes upward with the increase in c (see Figure 3.2). This is because increase in c leads to a decline in η , and therefore the term $\eta/\eta - 1$ in equation (4) increases as β is a constant. Turning to equation 5, the graph of this equation (PQ curve) slopes downward with the increase in c (this is firm's average cost curve).